

Eupatorium purpureum
(Asteraceae)

Gravel Root,

Joe Pye Weed

Description Erect perennial growing to 5 ft (1.5 m). Has whorls of pointed oblong leaves and clusters of purple-pink florets.

Habitat & Cultivation Gravel root is native to eastern North America. The root is unearthed in autumn.

Part Used Root.

Constituents Gravel root contains a volatile oil, flavonoids, and resin.

History & Folklore The plant's alternative name, Joe Pye weed, is in honor of the Native American said to have used it to cure New Englanders of typhus. Native Americans used the herb as a diuretic and as a remedy for genitourinary conditions. The root was listed in the *Pharmacopoeia of the United States* from 1820 to 1842.



Gravel root is especially helpful for urinary tract problems.

Medicinal Actions & Uses As its common name indicates, gravel root is a valuable herb for urinary tract problems. It helps to prevent the formation of kidney and bladder stones and may diminish existing stones. Gravel root is also useful for cystitis, urethritis, prostate enlargement (and other forms of obstruction), and for rheumatism and gout. The root is thought to help the latter two conditions by increasing the removal of waste by the kidneys.

Related Species *E. maculatum*, native to eastern North America, is used to treat kidney and urinary problems. See also *E. cannabinum*, p. 208, and *E. perfoliatum*, preceding entry.

Euphorbia hirta syn. *E. pilulifera*
(Euphorbiaceae)

Pill-bearing Spurge,

Asthma Plant

Description Erect annual or perennial plant growing to 20 in (50 cm), with pointed oval leaves and clusters of small flowers.

Habitat & Cultivation Native to India and Australia, pill-bearing spurge is now widespread throughout the tropics. The aerial parts of the plant are gathered when it is in flower.

Parts Used Aerial parts.



Pill-bearing spurge is recommended as a treatment for asthma.

Constituents Pill-bearing spurge contains flavonoids, terpenoids, alkanes, phenolic acids, shikimic acid, and choline. The latter two constituents may be partly responsible for the antispasmodic action of this plant.

History & Folklore As its name suggests, this plant was traditionally used in Asia to treat asthma.

Medicinal Actions & Uses A specific treatment for bronchial asthma, pill-bearing spurge relaxes the bronchial tubes and eases breathing. Mildly sedative and expectorant, it is also taken for bronchitis and other respiratory tract conditions. It is most often used along with other anti-asthmatic herbs, notably gumplant (*Grindelia camporum*, p. 218) and lobelia (*Lobelia inflata*, p. 110). In the Anglo-American tradition, pill-bearing spurge is taken to treat intestinal amebiasis.

Related Species The Cherokee used *E. maculata* to treat sore nipples and skin disorders. Many other North American *Euphorbia* species were used for constipation. A decoction of *E. lancifolia*, native to the West Indies, is used to stimulate breast-milk production. *E. atoto* is used in Malaysia and Indochina to induce a delayed period and as an abortifacient. Many species of *Euphorbia* are used as arrow poisons.

⚠ **Caution** Take pill-bearing spurge only under professional supervision.

Euphorbia pekinensis
(Euphorbiaceae)

Da Ji

Description Erect annual or perennial plant with oblong leaves and dense clusters of small flowers.

Habitat & Cultivation Native to China, *da ji* is cultivated mainly in the eastern and central provinces. The root is unearthed in early spring.

Part Used Root.

Constituents *Da ji* contains euphorbon.

Medicinal Actions & Uses *Da ji* is classified as a toxic herb in Chinese herbal medicine, and therefore it is prescribed only for relatively serious illnesses. It is taken as a cathartic to purge excess fluid in conditions such as pleurisy and ascites (excess fluid in the abdomen), and for the treatment of kidney problems, especially nephritis. *Da ji* is applied externally to inflamed sores to reduce swelling.

Research Research in China indicates that *da ji* is therapeutically useful in the treatment of ascites and nephritis; however, it produces significant side effects.

Related Species *E. kansui* is similar to *da ji* but has a stronger cathartic action. In Chinese herbal medicine it is only given to people with strong constitutions.

⚠ **Caution** *Da ji* is a toxic plant. Take only under professional supervision.

Euphrasia spp. (Scrophulariaceae) Eyebright

Description Creeping semi-parasitic annual growing to 20 in (50 cm). Has tiny oval leaves and small scallop-edged white flowers with yellow spots and a black center; somewhat resembling an eye.

Habitat & Cultivation Common in Europe, eyebright thrives in meadows and open grassland. It is gathered in summer when in flower.

Parts Used Aerial parts.

Constituents Eyebright contains iridoid glycosides (including aucubin), flavonoids, tannins, lignans, and phenolic acids. Aucubin has antibacterial, anti-inflammatory, and antispasmodic activity.

History & Folklore Eyebright's use for eye problems was due in part to the Doctrine of Signatures, an ancient belief that the way that a plant looks—its "sign"—indicates the ailments for which it should be used as a medicine.

Medicinal Actions & Uses Eyebright relieves inflammation and tightens mucous membranes and finds specific use in the treatment of conjunctivitis and blepharitis. Its ability to counter mucus means that it is often used for infectious